

Pass thru: 70 uA

The schematic diagram shows an I2C bus with four components: H1 M3, H2 M3, H3 M3, and H4 M3. The bus is connected to GND. Below the bus, four footprints are shown: FID1, FID2, FID3, and FID4, all labeled as Fiducial.

liion

sw_source

File: liion.kicad_sch

File: sw_source.kicad_sch

Schematic diagram of the FT2600Q-U USB-to-UART bridge module. The central component is the FT2600Q-U (U3), a yellow rectangular chip. It has various pins connected to external components. On the left, there are two LEDs: a red LED (D1) and a green LED (D2), both connected to +3.3V and ground. Below them are two resistors, R16 (270R) and R22 (68R), connected to +3.3V and ground. The UART_TX_LED and UART_RX_LED pins are connected to these LEDs. The module also has a debugger interface with pins 1 through 7 connected to +3.3V and ground. The FT2600Q-U has pins for #RTS, #CTS, USB_RXD, USB_TXD, SCL_USB, SDA_USB, #ENUM_FTDI_USB, #RTS, #CTS, USB_RXD, USB_TXD, SCL_USB, SDA_USB, #ENUM_FTDI_USB. On the right, there are two USB connectors (R33 and R34) connected to USB_FTDI_D+ and USB_FTDI_D-. There is also a reset button (JP3) connected to RST# and a 100R resistor (R38). The module is powered by +3.3V and ground. The schematic is labeled "Suspend: 356 uA".

Je zde I2C master switch, tím by mělo být složeno by komponenty byly "programovatelné" přímou přes USB pomocí FT232RL, díky tomu lze USB-USB+UART a zároveň USB-I2C bridge, díky tomu by to mělo zanechat konfiguraci.

Operating mode: 200 uA
Standby mode: 80 uA

U4 I2C: 0x70

PCA9541ABS/01

0x70

VSS_psd

VSS

A0

A1

A2

A3

GND

SCL_MCU

SDA_MCU

SCL_USB

SDA_USB

INT0

SCL_MST0

SDA_MST0

INT1

SCL_MST1

SDA_MST1

RESET

VDD

INT_IN

SDA_SLAVE

SCL_SLAVE

R21

R23

R24

R25

R36

R37

TP2

R30

+3.3V

+3.3V

+3.3V

C8

100n

GND

Active: 1.5mA
Standby: 6 uA

+3.3V
I2C: 0x44
U6 0x44
SHT31-DIS
ADDR
RESET
R
VSS
VDD
SDA
SCL
ALERT
GND

+3.3V
C16 100n

Active: 12.5 uA
Standby: 0.14 uA
U9 I2C: 0x76
MS561101BA03-50
VDD
PS
GND
CSB_1
SCLK
SDI/SDA
SDO
CSB_2
GND

+3.3V
C18 100n

AB507-120-32.768KHZ-T
Y3
OSC1
OSC0
U10
PCF85263A
SCL
SDA
VBAT
GND
INTA
CLK
TS
GND

+3.3V
C17 100n

EXT_DETECTION_CHARGER

+3.3V
I2C: 0x50
0x58
AT24C564-SSHM
U5
A0
A1
A2
GND
VCC
SDA
SCL
WP
GND
GND
GND

+3.3V
C50 100n

Active: 1.0 mA
Standby: 2 uA

+3.3V
C20 100n

Active: 1.0 mA
Standby: 2 uA

I2C addresses
=====

0x44: HYGROMETER
0x76: ALTIMET
0x70: I2C switch
0x6B: Charger
0x55: Gauge
0x50: EEPROM
0x58: EEPROM SN
0x51: RTC
0x71: SDreader

+BATT
R41 100R
JP4
C20 100n
GND

R45 10K
R82 100R
BTN_USER_A

Operating mode: ~7 mA

Active: 4.5 mA
Shutdown: 30 uA

Enabled by default (EN – high)

The diagram shows a yellow rectangular component labeled "MOLEX 46436-9327" with a connector symbol "J4".

Left Side Pins (A1-B6):

- A1, A2, A3, A4, A5, A6 (top row)
- B1, B2, B3, B4, B5, B6 (bottom row)

Right Side Pins (C1-D6):

- C1, C2, C3, C4, C5, C6 (top row)
- D1, D2, D3, D4, D5, D6 (bottom row)

Bottom Side Pins (E1-E6):

- E1, E2, E3, E4, E5, E6 (bottom row)

Connections:

- Power:** Pins A3, A4, B3, and B4 are connected to a 3V3_EXT and 5V_EXT supply. Pins P1, P2, P3, and P4 are connected to a +BATT supply.
- Signal:** Pins C1, C2, C3, C4, C5, and C6 are connected to SCL_OUT, SDA_OUT, RST#, SCLK_OUT, MISO_OUT, and MOSI_OUT respectively. Pins D1, D2, D3, D4, D5, and D6 are connected to EXT_A0, EXT_A1, EXT_A2, EXT_A3, EXT_A4, and EXT_A5 respectively. Pins E1 through E6 are connected to EXT_C0, EXT_C1, EXT_TX, EXT_RX, EXT_B0, and EXT_B1 respectively.

Active: 68 μ A
Disabled: 15 μ A

U7
TMUX1574DYYR

R39
10k

~~GND~~ ~~SPL_MUX_SEL~~ 1 SEL VDD 16 ~~+3.3V~~

MOSI_SD 2 S1A EN 15 ~~GND~~

MOSI_OUT 3 S1B S4A 14 ~~GND~~

MOSI_MCU 4 D1 S4B 13 ~~GND~~

MISO_SD 5 S2A D4 12 ~~GND~~

MISO_OUT 6 S2B S3A 11 SCLK_SD

MISO_MCU 7 D2 S3B 10 SCLK_OUT

~~GND~~ 8 GND D3 9 SCLK_MCU

+3.3V

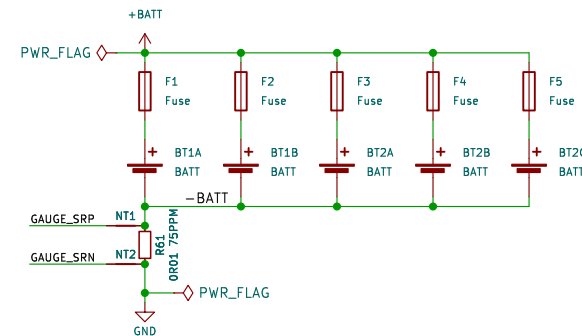
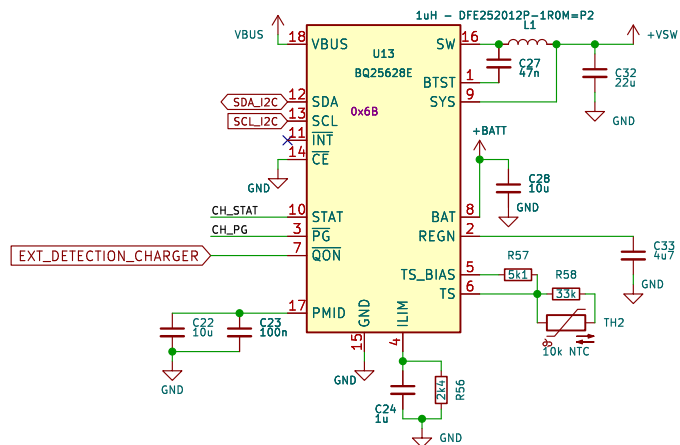
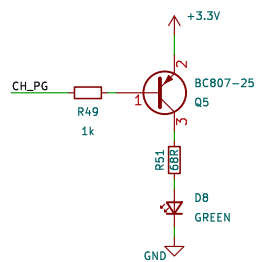
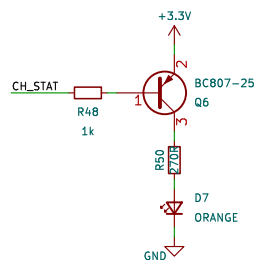
C19
1u

GND

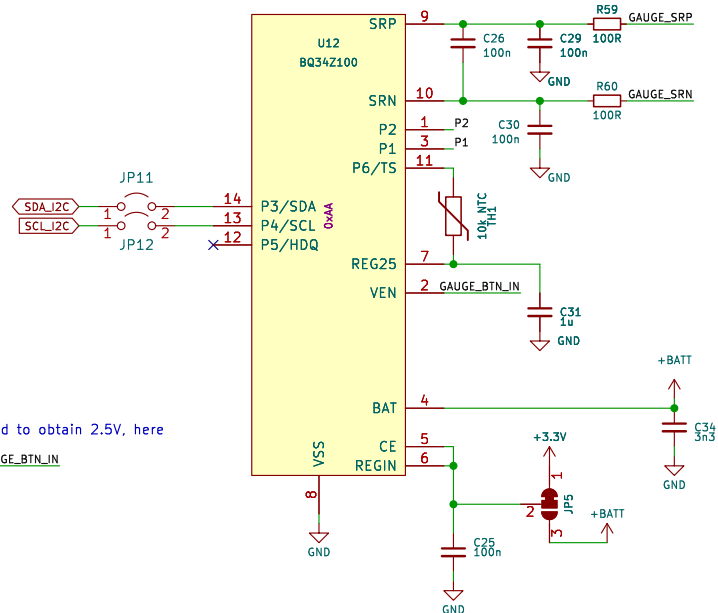
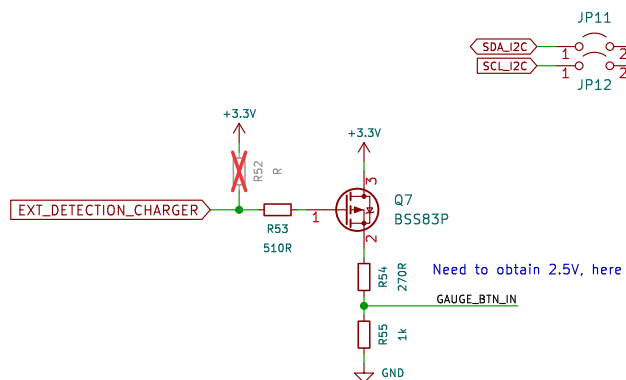
Size: A3	Page: 1/3	Date: 11-2023
----------	-----------	---------------



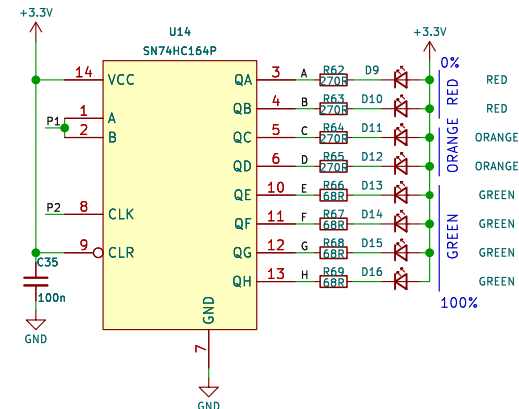
BATTERY CHARGER



FUEL GAUGE



Battery Status Indicator



Title: BATDATUNTIO1

Universal battery and data storage unit.

Author: MLAB.cz, roman-dvorak

Size: A4

Page: 2/3

Date:



